

Skill 12 os

SESSION 23

Task 1: To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance

Code:

```
rohita@DESKTOP-1UHU58C: ~  
GNU nano 8.7.1 s12T1.c *  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <sys/wait.h>  
#include <time.h>  
  
int main() {  
    int jobs = 3;  
    pid_t pid;  
    time_t start, end;  
  
    start = time(NULL);  
  
    for (int i = 1; i <= jobs; i++) {  
        pid = fork();  
  
        if (pid == 0) {  
            printf("Job %d started (PID: %d)\n", i, getpid());  
            sleep(i);  
            printf("Job %d completed\n", i);  
            exit(0);  
        }  
    }  
  
    for (int i = 0; i < jobs; i++)  
        wait(NULL);  
  
    end = time(NULL);  
  
    printf("All jobs completed successfully.\n");  
    printf("Total execution time: %ld seconds\n", end - start);  
  
    return 0;  
}
```

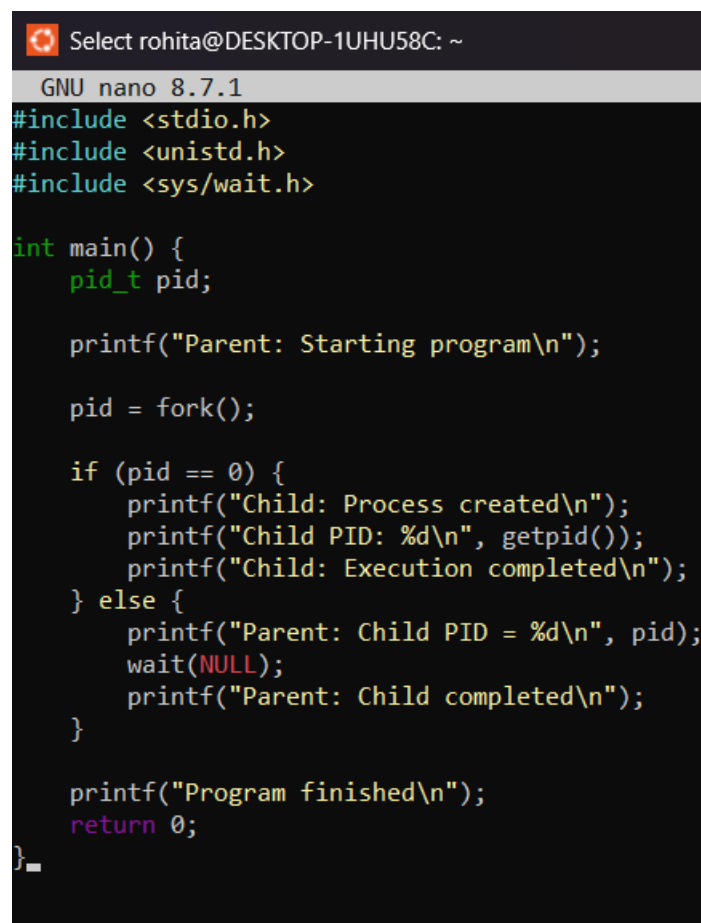
Output:

```
Job 1 started (PID: 9929)  
Job 2 started (PID: 9930)  
Job 3 started (PID: 9931)  
Job 1 completed  
Job 2 completed  
Job 3 completed  
All jobs completed successfully.  
Total execution time: 3 seconds
```

Observation: Multiple jobs were launched simultaneously using `fork()`. Their execution was monitored using `wait()`, and the total execution time was measured. All jobs completed successfully, demonstrating stable parallel process execution.

Task 2: To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentatio

Code:

A screenshot of a terminal window with a dark background. The title bar shows a red icon and the text "Select rohita@DESKTOP-1UHU58C: ~". Below the title bar, a grey bar indicates "GNU nano 8.7.1". The main area contains C code with syntax highlighting:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

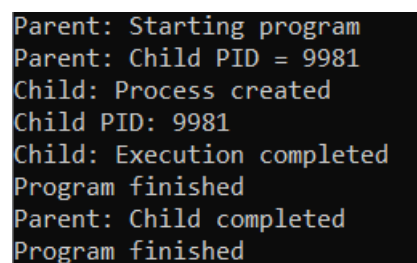
    printf("Parent: Starting program\n");

    pid = fork();

    if (pid == 0) {
        printf("Child: Process created\n");
        printf("Child PID: %d\n", getpid());
        printf("Child: Execution completed\n");
    } else {
        printf("Parent: Child PID = %d\n", pid);
        wait(NULL);
        printf("Parent: Child completed\n");
    }

    printf("Program finished\n");
    return 0;
}
```

Output:

A screenshot of a terminal window showing the output of the program. The text is as follows:

```
Parent: Starting program
Parent: Child PID = 9981
Child: Process created
Child PID: 9981
Child: Execution completed
Program finished
Parent: Child completed
Program finished
```

Observation: The program successfully created a child process using `fork()` and synchronized it using `wait()`. The execution flow and system calls were verified successfully.

SESSION 24

Task 1: To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables

Code:

```
rohita@DESKTOP-1UHU58C:~$ mkdir project
rohita@DESKTOP-1UHU58C:~$ cd project
rohita@DESKTOP-1UHU58C:~/project$ mkdir src docs
rohita@DESKTOP-1UHU58C:~/project$ src/main.c README.md Makefile
-bash: src/main.c: No such file or directory
rohita@DESKTOP-1UHU58C:~/project$ touch src/main.c README.md Makefile
rohita@DESKTOP-1UHU58C:~/project$ ls -R
```

Output:

```
.:
Makefile  README.md  docs  src

./docs:

./src:
main.c
rohita@DESKTOP-1UHU58C:~/project$ _
```

Observation: The project files were organized into proper directories and the repository structure was verified successfully.

Task 2: To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review

Code:

```
rohita@DESKTOP-1UHU58C: ~
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

void handler(int sig) {
    printf("Signal received successfully\n");
}

int main() {
    signal(SIGINT, handler);
    printf("Demo started\n");
    printf("Pipeline and signal features ready\n");
    sleep(2);
    printf("Final project review completed\n");
    return 0;
}
```

Output:

```
rohita@DESKTOP-1UHU58C:~$ ./a.out
Demo started
Pipeline and signal features ready
Final project review completed
rohita@DESKTOP-1UHU58C:~$
```

Observation: The demo features were executed successfully, and the project functionality was verified for final presentation.